

MADHAVAN A RAMANUJAN

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SUMMARY

Computer Science undergraduate specializing in AI & Robotics with hands-on experience designing, training, and deploying deep learning and computer vision systems. Skilled in end-to-end model development using PyTorch, TensorFlow, and OpenCV, with a strong background in data preprocessing pipelines, model optimization, and REST API deployment. Certified in Generative AI and Cloud Architecture, with a proven ability to translate theoretical algorithms into scalable, real-time automated applications.

EDUCATION

VIT Chennai | B.Tech, Computer Science & Engineering (AI & Robotics) | CGPA: 8.08 | 2023–2027

TECHNICAL SKILLS

Languages: Java, Python, C, C++, SQL

Core CS: Data Structures & Algorithms, Operating Systems, DBMS (MySQL), Computer Networks, OOP

Machine Learning: TensorFlow, Keras, CNNs, Transfer Learning, OpenCV, Computer Vision, OCR, LLM APIs, Prompt Engineering

Data Analytics Tools: Power BI, MS Excel, MySQL

Backend & Web: Flask, REST APIs, HTML/CSS, Docker

Tools: Git, GitHub, Jupyter Notebook, Arduino

CERTIFICATIONS

- Oracle Cloud Infrastructure AI Foundations Associate
- Oracle Cloud Infrastructure Generative AI Professional

PROJECTS

AI-Powered Dyslexia Detection System (Python, TensorFlow, Flask, OCR, OpenCV)

github.com/Maddy152006/dyslexia-detection-ai

- Architected a CNN-based character classification pipeline trained on 100,000+ handwritten glyph samples to detect dyslexic writing patterns across Normal, Reversal, and Corrected classes with ~96–97% validation accuracy
- Built a Flask-based OCR inference system integrating PyTesseract, OpenCV preprocessing, adaptive thresholding, and character segmentation for real-time dyslexia screening and annotation
- Implemented dataset augmentation, class rebalancing, confidence-based prediction logic, and confusion matrix evaluation using TensorFlow/Keras for explainable character-level analysis

Vision-Guided Robotic Waste Segregation System (Python, TensorFlow, OpenCV, MobileNetV2, Arduino Uno)

github.com/Maddy152006/waste-segregation-ai

- Developed a transfer learning-based waste classification pipeline using MobileNetV2 to perform multi-class categorization across Metal, Paper/Cardboard, Plastic, and Trash classes with ~88% validation accuracy
- Built a real-time computer vision inference system integrating OpenCV, DroidCam webcam streaming, frame preprocessing, and live prediction overlays for automated waste detection and classification
- Integrated TensorFlow-based prediction outputs with an Arduino Uno-controlled robotic arm via serial communication, enabling calibrated servo-based waste segregation and category-specific routing
- Applied dataset augmentation, transfer learning fine-tuning, and confusion matrix evaluation pipelines using TensorFlow/Keras to improve classification generalization for smart recycling automation

Corporate Fraud & Financial Risk Audit Pipeline (SQL, Python, Power BI, Advanced DBMS)

github.com/Maddy152006/Financial-Fraud-Audit-Data-Pipeline/tree/main

- Engineered a robust relational data warehouse database schema using MySQL to ingest and evaluate high-velocity ledger transactions against user profile entities
- Authored forensic query detection heuristics using **CTEs** and **window functions** (LAG) to systematically isolate stolen card velocity spikes, cross-border geographic mismatches, and money laundering structuring patterns
- Developed optimized relational database reporting views utilizing group aggregations to resolve presentation-layer data loops, cleanly exposing 880+ true risk logs to an interactive **Power BI dashboard**

AI Meeting Summarizer Pipeline (Python, Streamlit, Whisper ASR, LLMs, APIs)

github.com/Maddy152006/meeting-summarizer

- Engineered an automated transcription and summarization web application with a modular backend architecture
- Integrated Groq's Whisper-large-v3 API for highly accurate verbatim speech-to-text conversion of meeting audio
- Leveraged a 120B parameter LLM with optimized prompts to dynamically extract executive summaries, key decisions, and structured action-item tables with deadlines

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- **Competitive Badminton Player** — Won multiple junior badminton tournaments and competed in Karnataka State Open and state ranking tournaments, demonstrating discipline, consistency, and competitive spirit
- Member, VIT Chennai Sports Club (2024–2025)
- Participated in the VIT Chennai 5K Run